

**What are some
'out of the box'
or radical ideas for
the project's solution
space, informed by
your group's data
analysis?**



Week 7 Lecture

Idea Generation

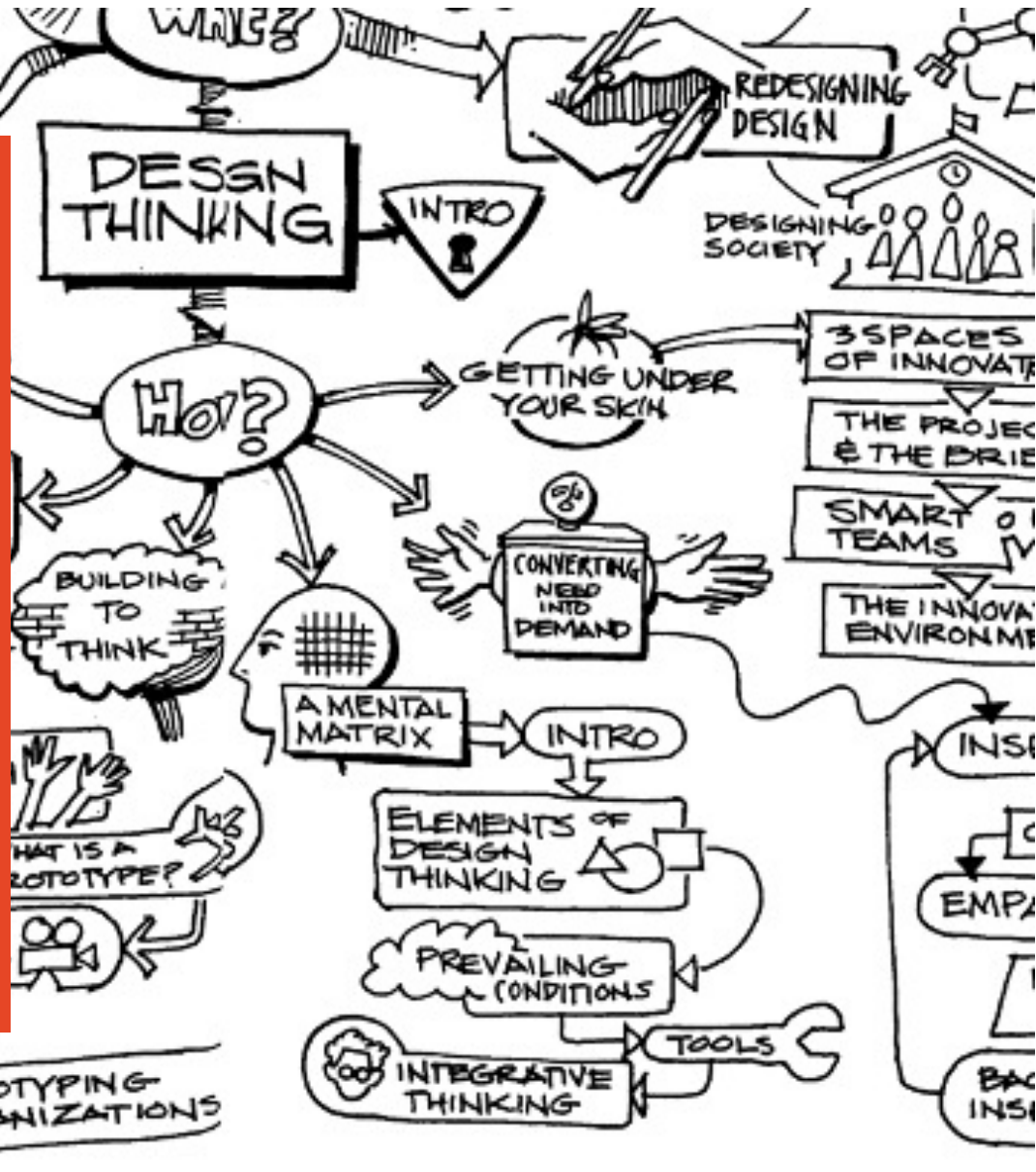
Lo-fi Prototyping

IDEA9106 Design Thinking
Semester 2 2026



THE UNIVERSITY OF
SYDNEY

CRTCOS 00026A



We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.



THE UNIVERSITY OF
SYDNEY

Overview: Week 7

- Design Thinking concepts & theory
- Design Thinking methods & practice
- Wrap Up

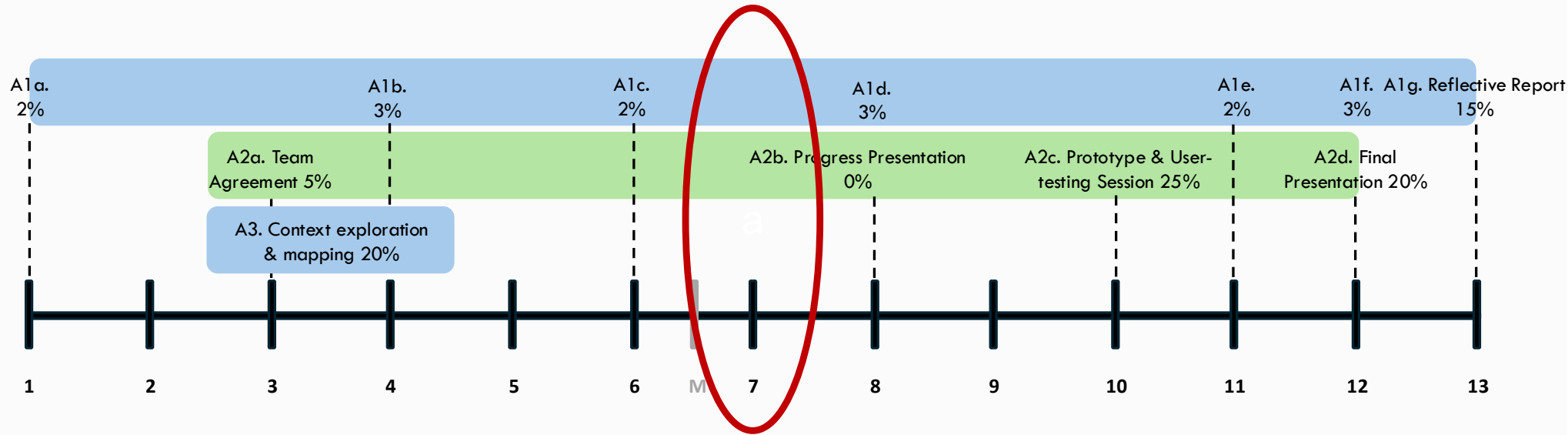


IDEA9106 Assessments Semester 2 2026

All assessments are mandatory

Questionati is the unit's AI for 24/7 answers to assessment questions

Assessments in this unit are connected, building towards A1g. And A2d.



MS: Mindset

PE: Peer & Self evaluation

Individual tasks

Teamwork tasks

A1: Reflective practice

A2: Collaborative learning

A3: Framing and Reframing

Weekly lecture attendance check

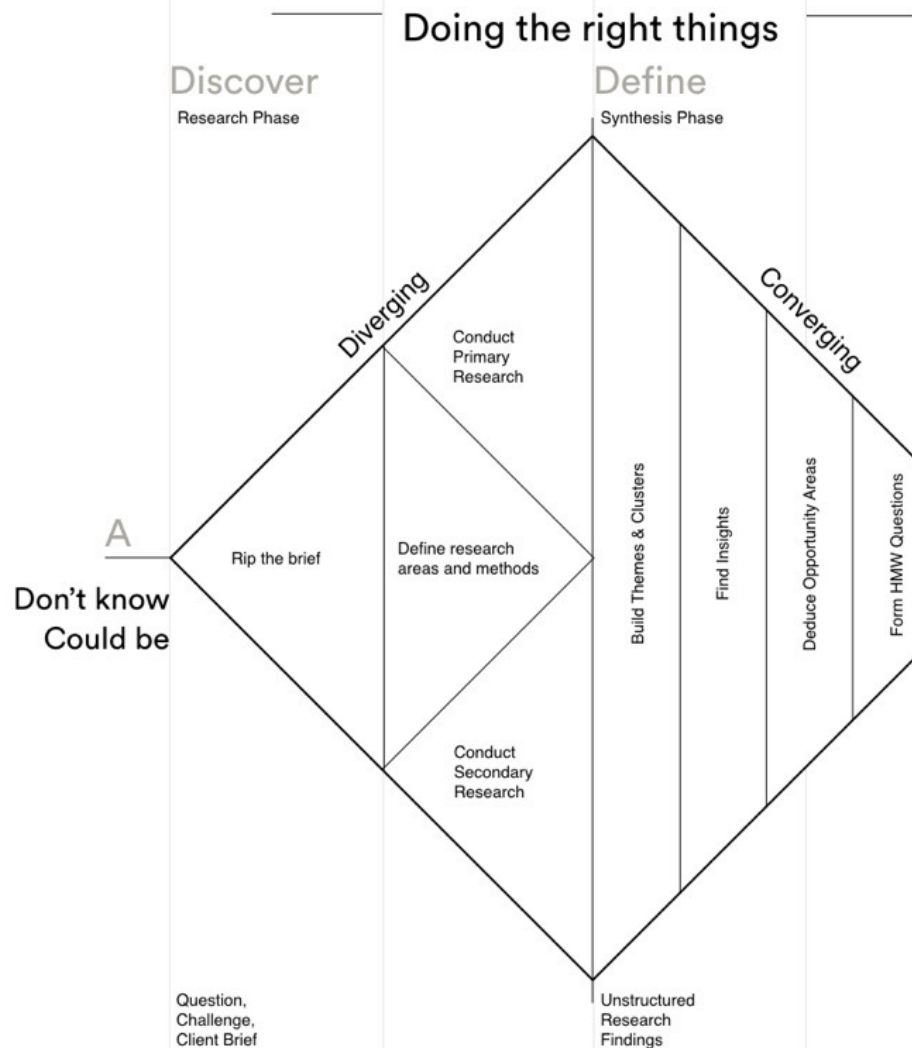
- In-person and online attendance, and recorded lecture views are monitored
- Students who attend or watch lecture recordings have a much better understanding of key concepts, and tutorial and assessment requirements
- No mark penalty but you will receive reminder emails about the importance of lectures



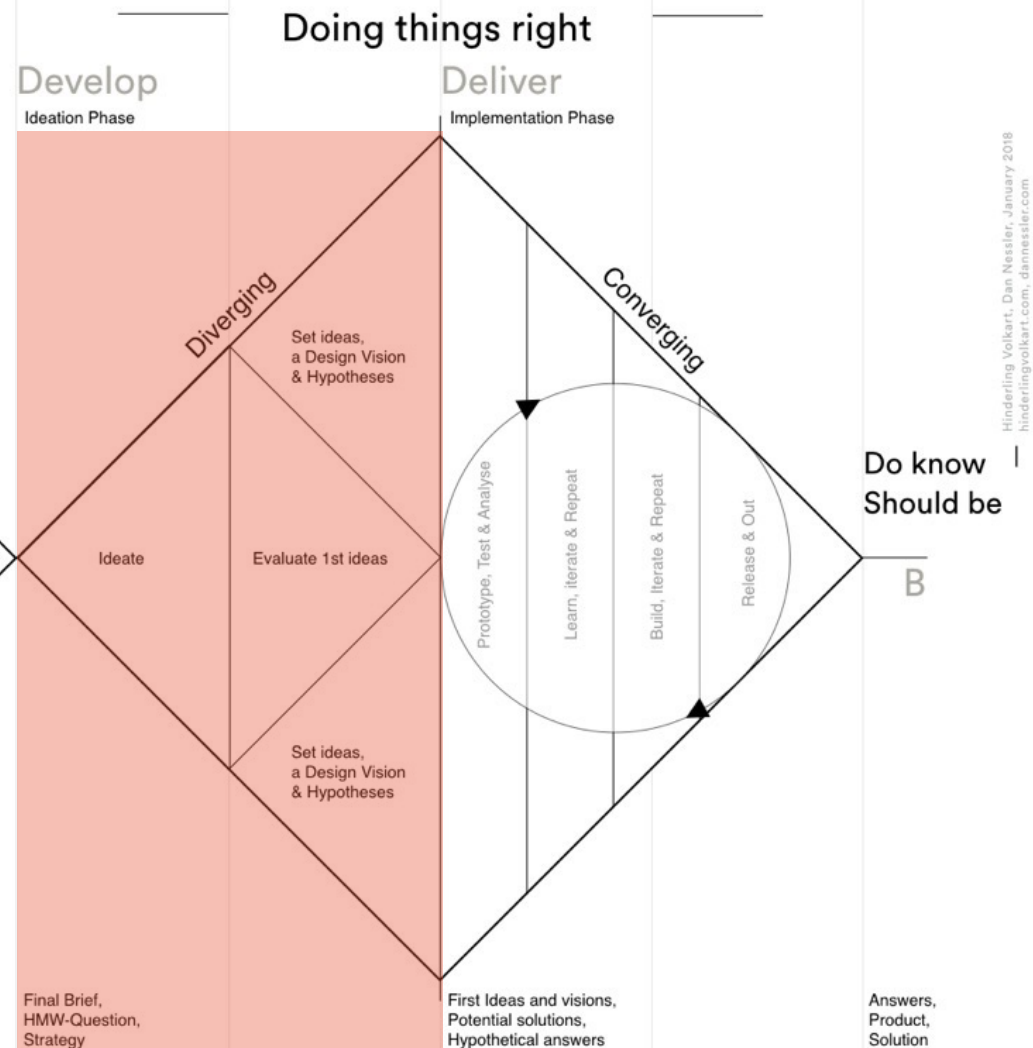
Overview: Week 7

- **Design Thinking** concepts & theory
- **Design Thinking** methods & practice
- **Wrap Up**





Revamped Double Diamond



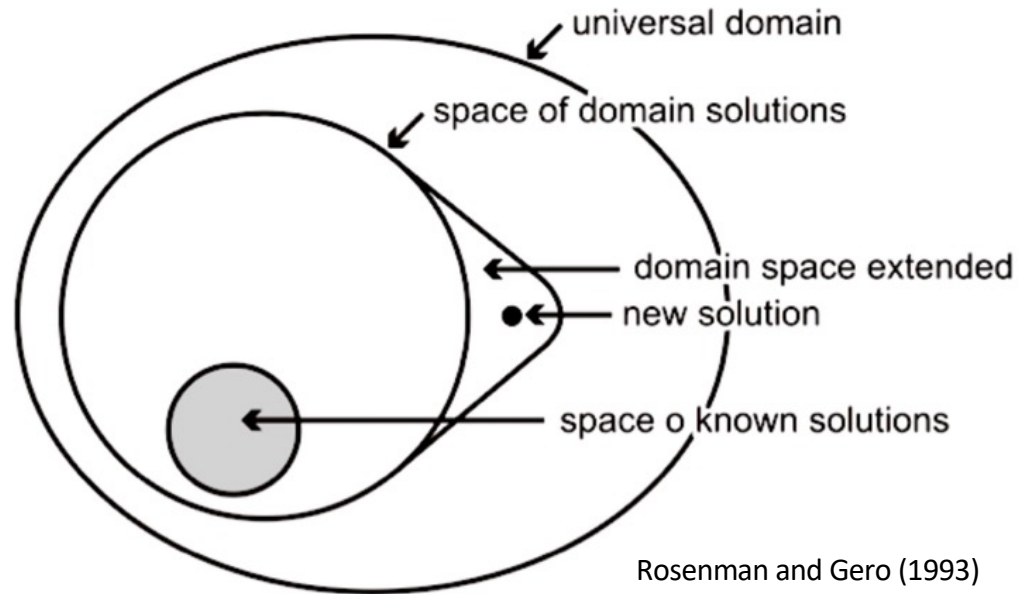
Idea generation
opening up the solution space
avoiding fixation
& sitting with uncertainty
iterating solutions
& redefining the problem

What is a solution space?

The solution space contains the set of possible answers to your brief, not the one you arrive at first.

What is a solution space?

The solution space contains the set of possible answers to your brief, not the one you arrive at first.



Rosenman and Gero (1993)

Expand the solution space

"The best way to have a good idea is to have many ideas."

- Linus Pauling

Dos and Don'ts for Better Ideation



Maintain parallel lines of thought

- Keep 2–3 idea pools alive;
- Develop them independently with intent.
- Merge and mix towards the end.



👉 *This week, you need to decide on two general concepts you would like to work on as a team.*

Go beyond core functions



- What your design does is not all it is.
- Ask about experience, ritual, emotion, status, failure.

📱 no-one buys a phone just because it makes phone calls.
It's all the additional features that make solutions valuable.

Build on others' ideas

- "Yes, *and...*" not "No, *but...*"
- Piggyback on their ideas instead of judging them.



Junior designers default to critique, because critique feels clever.

Make ideas visible

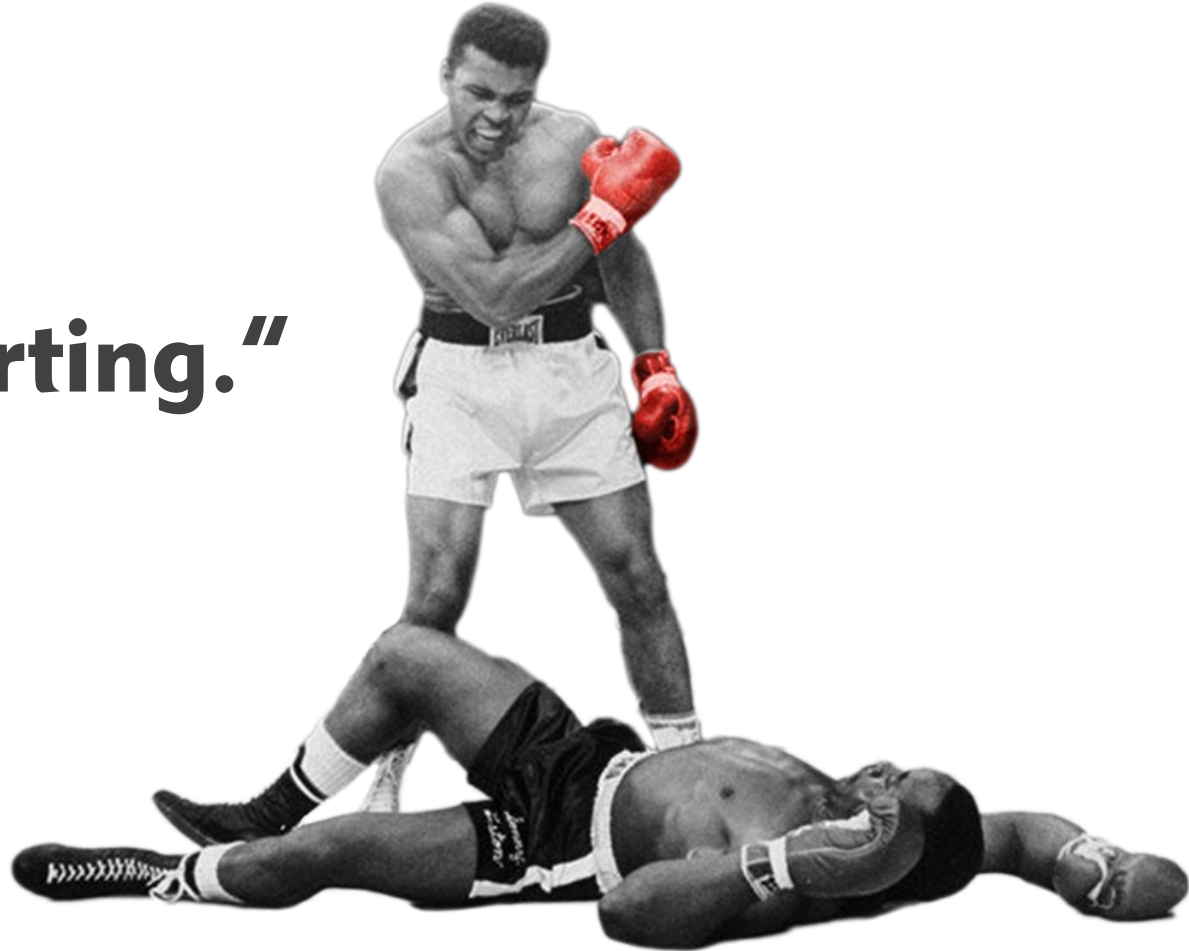


- Sketch, don't describe.
- Three rough pictures beat one tidy paragraph.

📖 This is why your journal exists. It is not a diary. It is the place your thinking becomes visible enough to be worked on.

**"I start counting
when it starts hurting."**

- Muhammad Ali



Don't dismiss small ideas



- Park them, don't bin them.
- The trivial idea is often the start of a real one.

📌 Half the time, the idea that eventually works is a grown-up version of one you almost threw away.

Don't fixate!



- Your first idea is usually your most familiar idea.

🧠 Once you've locked onto a solution, your brain quietly filters out alternatives, even when you think you're being open.

Don't rationalise too early

- If you explain an idea before you've explored it, you'll defend it instead of testing it.



💡 A new idea is very fragile. The worst thing you can do at that moment is build an argument for it.

Two modes of ideation.

- Breadth → generate range
- Depth → develop what's promising

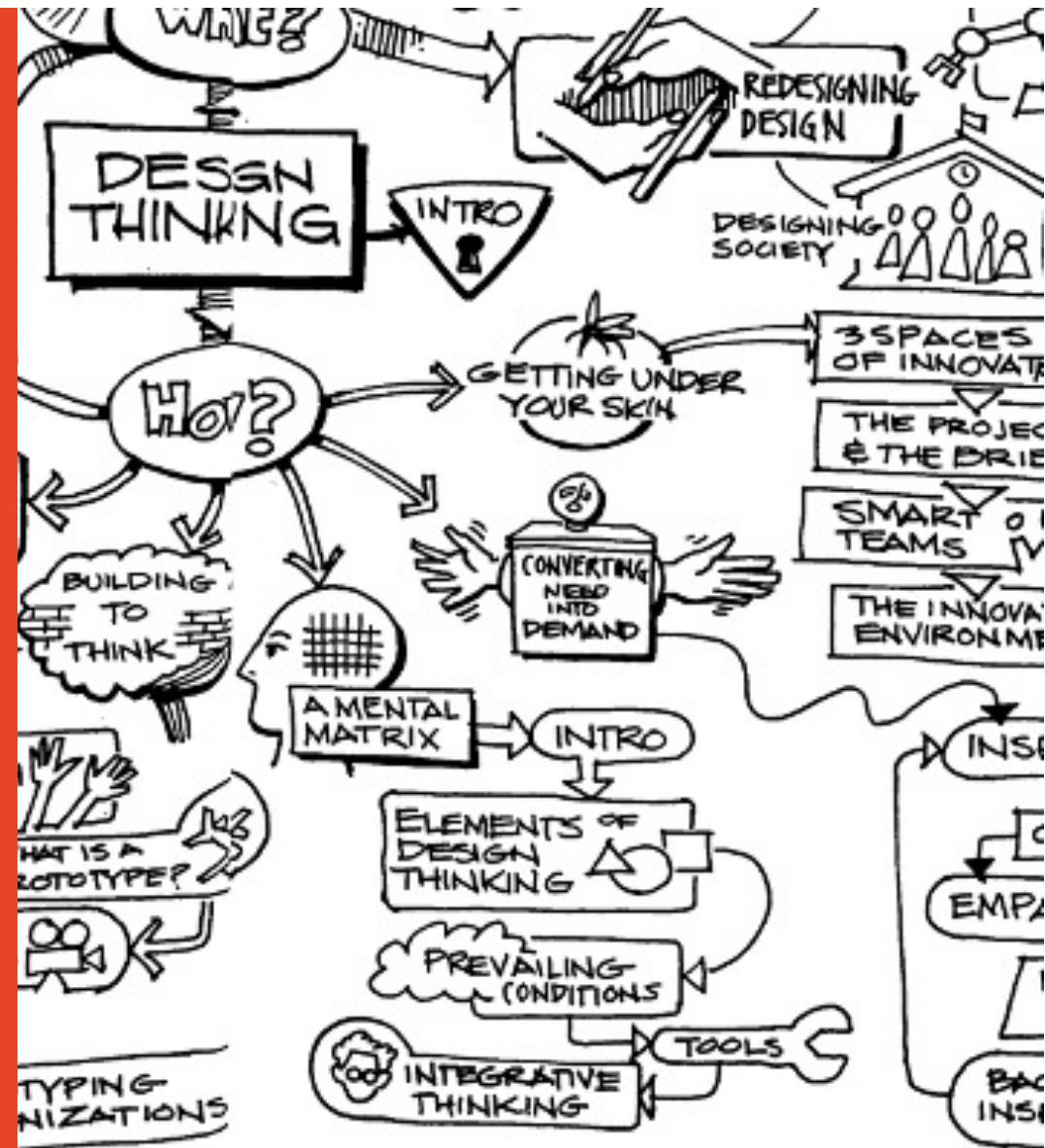
You need both.



In first mode, you're trying to open the space as wide as you can.
In second mode, you're taking a handful of directions and finding out what they actually look like when you push on them.

Overview: Week 7

- Design Thinking concepts & theory
- Design Thinking methods & practice
- Wrap Up



Idea generation methods

- Crazy 8s
- Brainstorming
- KJ Brainstorming
- Brainwriting
- Bodystorming
- Mindmapping
- Worst possible idea

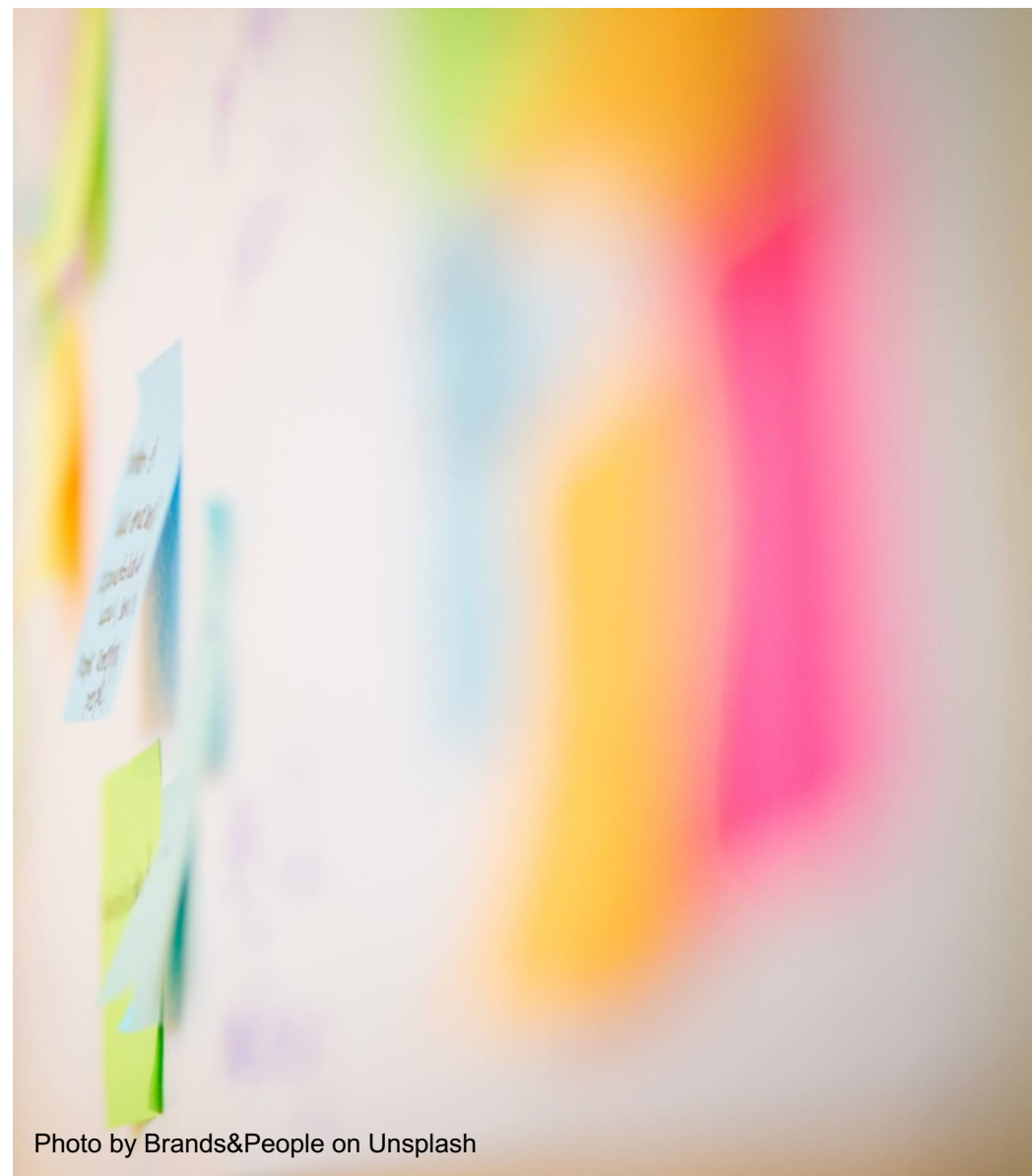
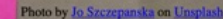


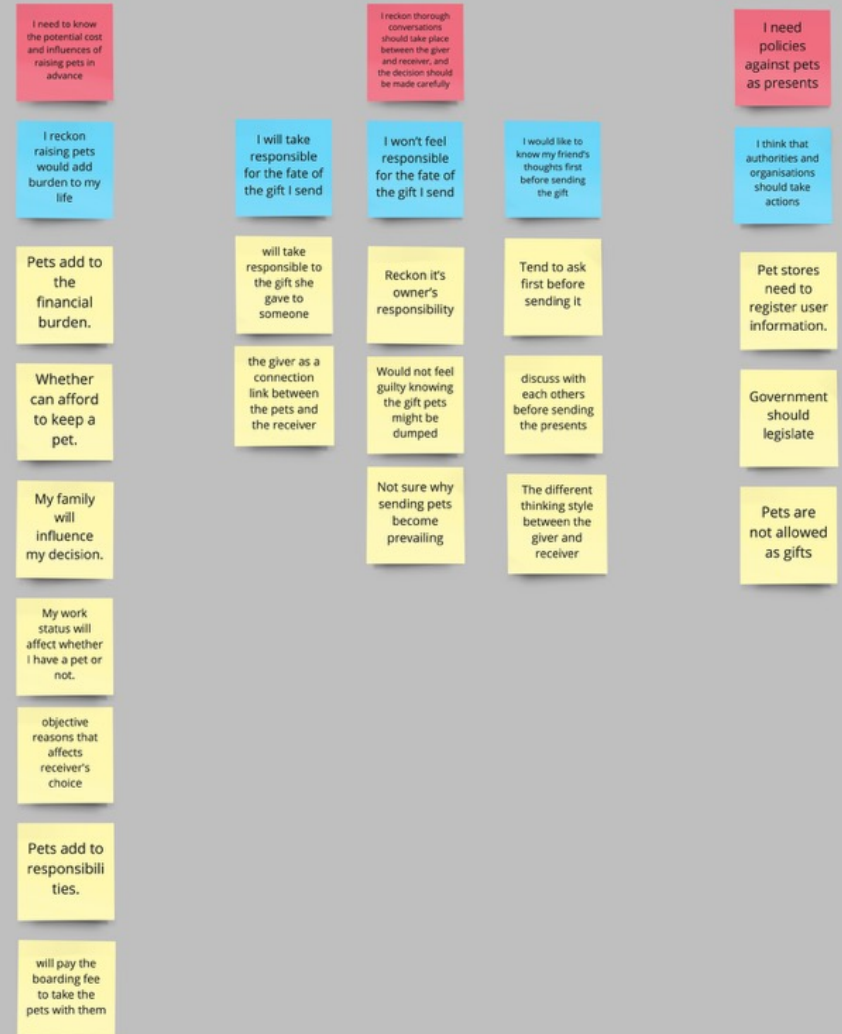
Photo by Brands&People on Unsplash

- Encourages people to produce thoughts and ideas, no matter how weird
- Any idea accepted
- Avoid criticizing or rewarding ideas
- Build off other's ideas
- Evaluate ideas at the end



KJ Brainstorming*

- Generate ideas individually
- Read notes, add new insights
- Cluster similar items
- Label each cluster
- Vote for top three clusters
- Discuss the emerging themes and their relevance for the project



Brainwriting 6-3-5*

- a form brainstorming
- balancing group dynamics
eg. shy and/or dominating personalities
- started with a discussion of the problem area
- six team members record three ideas in five-minute cycles



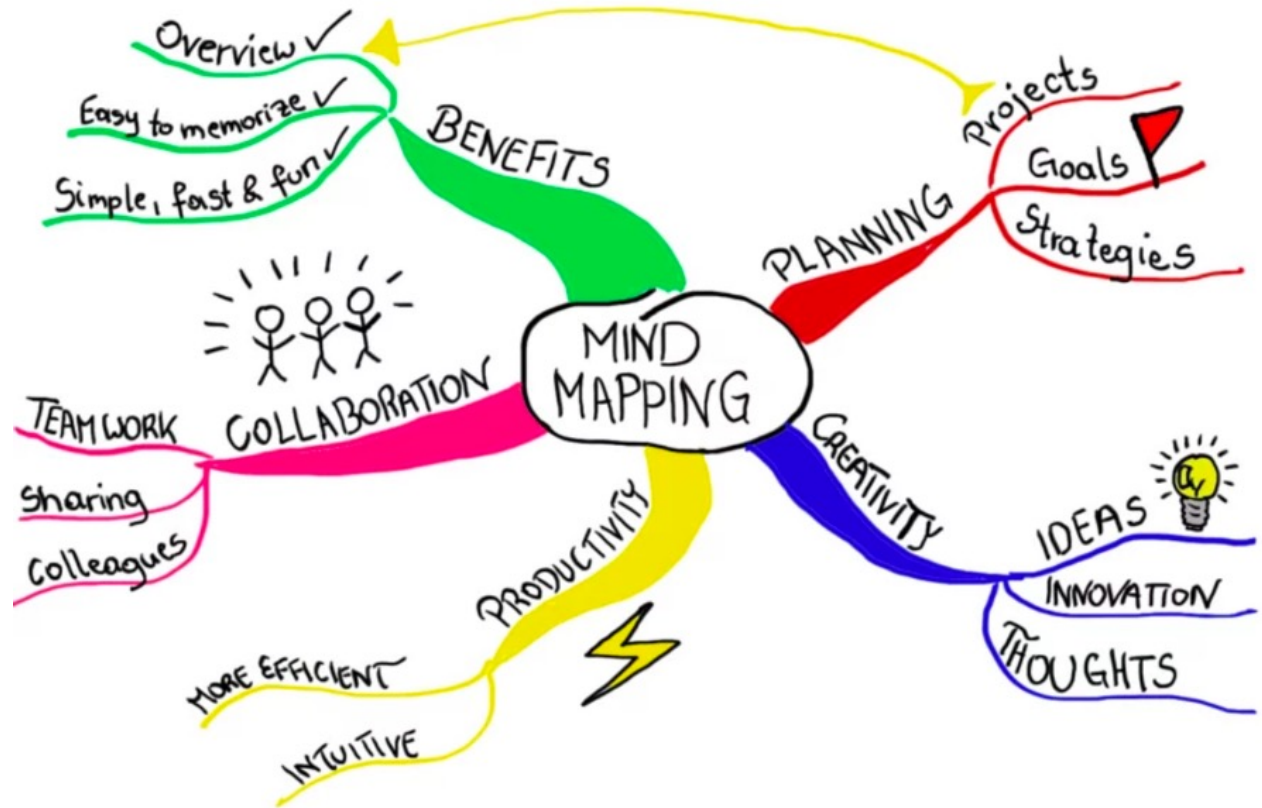
Bodystorming*

- generating ideas and insights through physical exploration.
- draws on techniques from theatre and improvisation
- active physical participation within particular scenarios and situations
- using props and role-play
- act out situations collaboratively



Mindmapping*

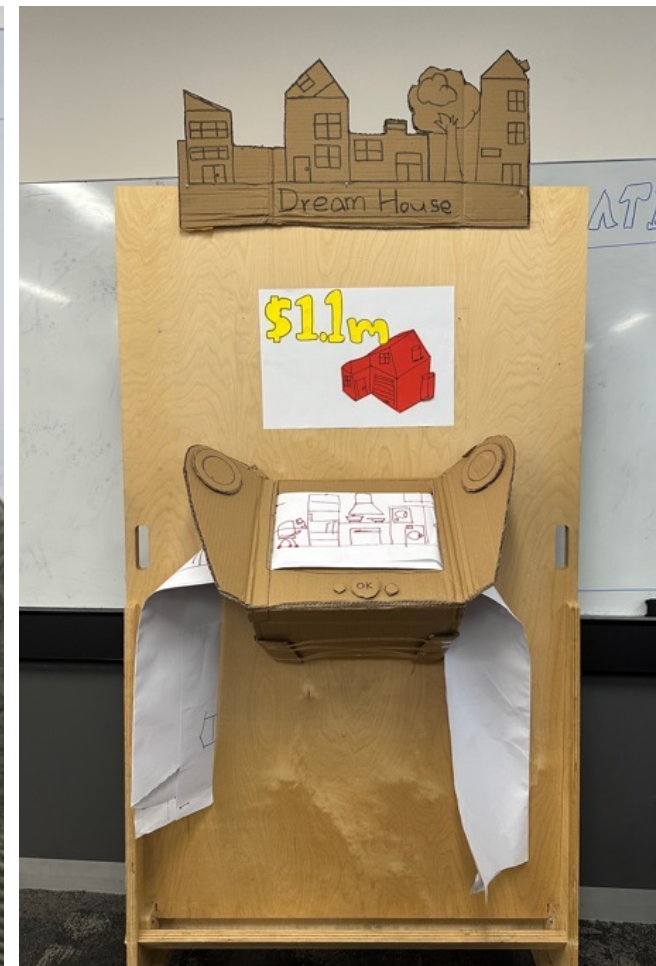
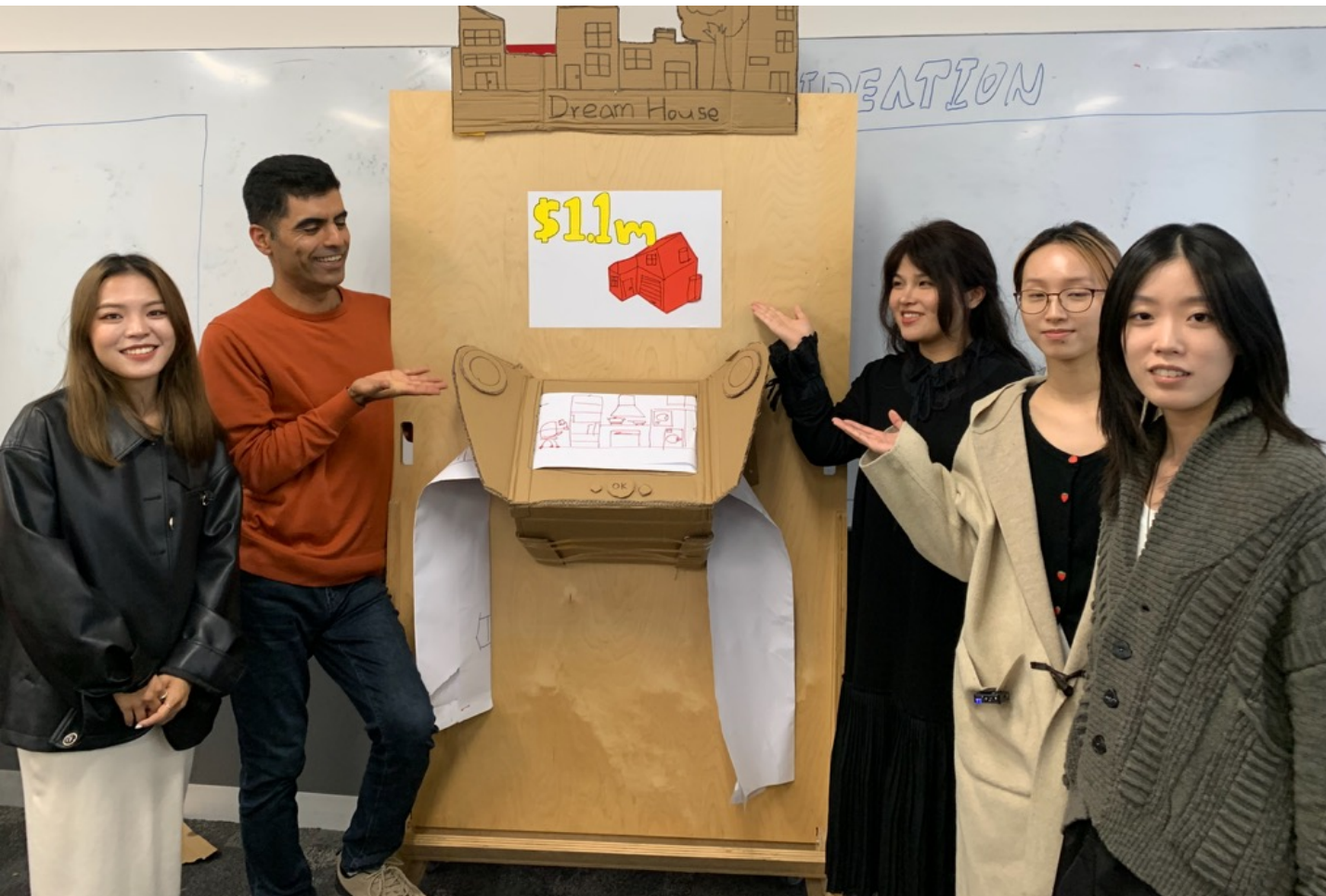
- method emerged from cognitive research
- good for dealing with large amounts of information
- WWWWWH Framework
Who, What, When, Why, Where & How
- get existing information out of your brain and onto the page



	Doing things right	
--	--------------------	--



Prototyping



A prototype is ...

- a first preliminary version from which other forms are developed
- an experimental, usually incomplete, version of a design
- focused on one aspect rather than the full design
- often done at scale
- used to not only to test products, but all kinds of things
- able to question and test a design concept

Why do designers prototype?

1

Reflect in/on action:

Get the idea out of your head so you can see it from the outside and notice gaps, biases, and opportunities.

2

Communicate:

Make the invisible shareable so teammates, stakeholders can see, question, and build on it.

3

Test & evaluate:

Put a prototype in front of people to probe usability, comprehension, and value.

4

Worldise the idea:

Let the concept take on the flavour of the real world: Let the context and constraints shape it.

Lo-fi prototypes

- Allow for quick exploration of ideas early in a design process
- Help facilitate reflection and discussion of potential design solutions within a team
- Enable feedback from prospective users through usability testing
- Constructed in a different medium to the final outcome
eg. paper, cardboard
- Quick, cheap and easy to make and change
- Incorporate 'ready-made objects', eg. post-its for buttons

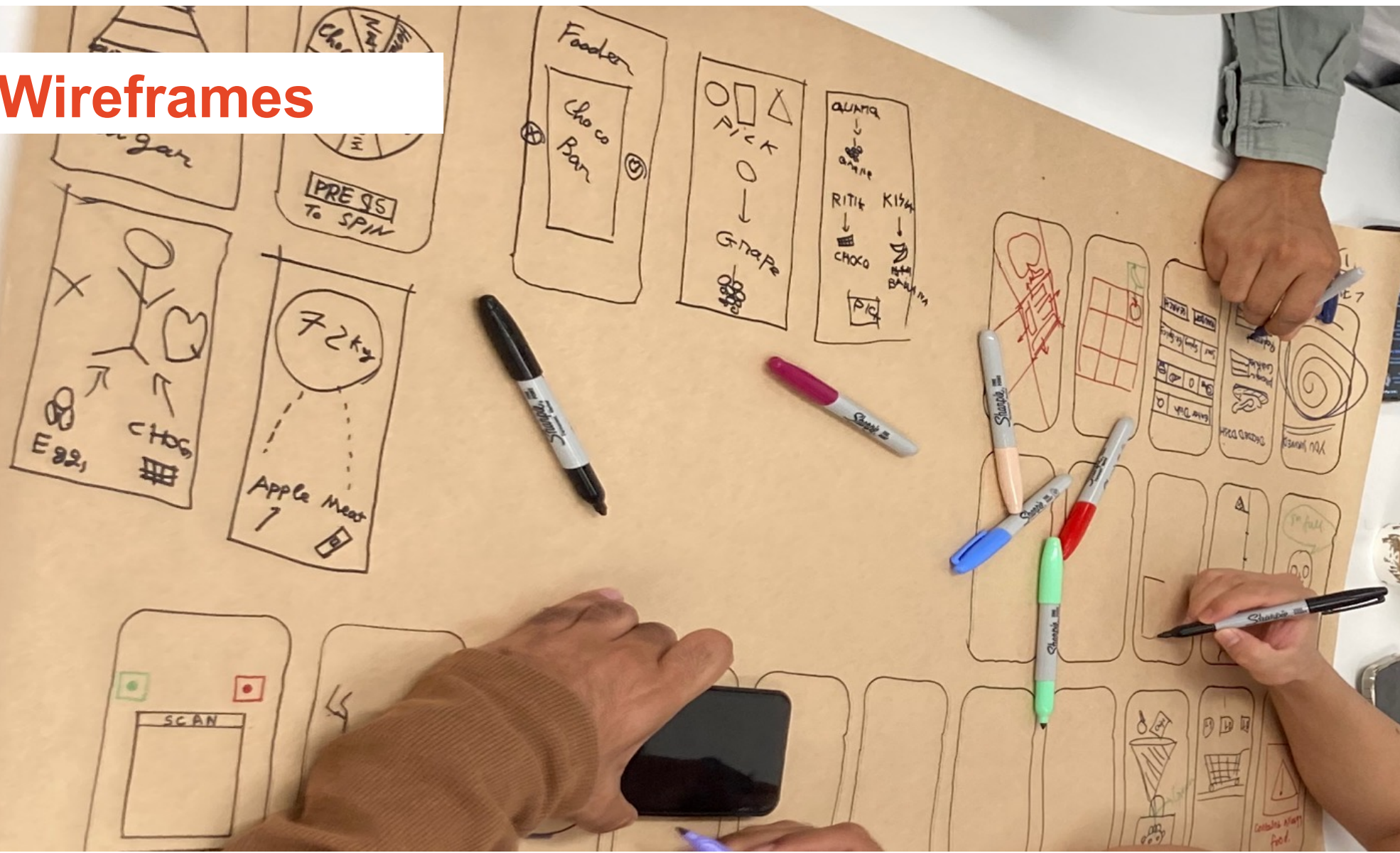
Paper prototypes



Mock-ups



Wireframes



Experience prototypes



CARDBOARD PROTOTYPING CASE STUDY

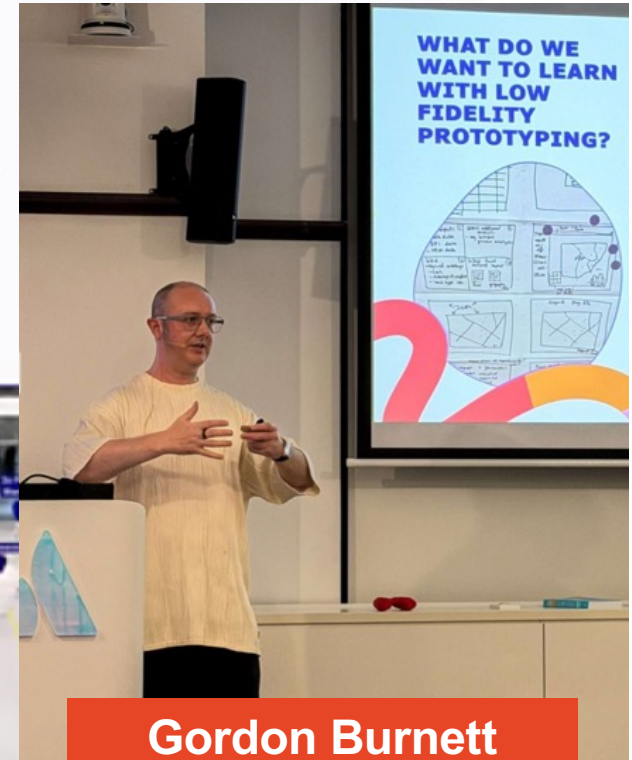
- Who:** Hospital Kitchen Redesign
- The Challenge:** New legislation on food waste and sustainable cleaning of equipment meant many hospital kitchens had to redesign their layout and workflows.
- Outcome:** Mapping out the plans at 1:1 scale and trying several different equipment layouts allowed the team to catch several critical OH&S issues quickly and then refine the spaces within the kitchen for optimised workflows.



KITCHEN PLANNING AND REDEVELOPMENT



LABELLED DRONE PHOTOS OF LAYOUT



Gordon Burnett
Health Prototyping
Centre
NSW eHealth
(Sydney Design Thinking Meetup)

Overview: Week 7

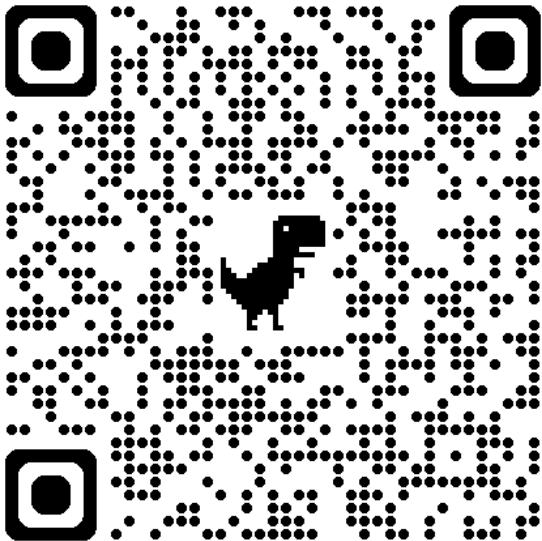
- Design Thinking concepts & theory
- Design Thinking methods & practice
- **Wrap Up**



AI and Design Practice

with *Thoughtworks*

DESIGN
@DUSK



Guests:

- Kate Linton (Design Principal)
- Katie Todeschini-Martin (Lead Product Manager)
- Neha Madan (Lead Consultant)
- Anh Le (Lead Experience Designer).

Host: Morteza Pourmohamadi

When

6 – 8pm

Thursday 24 September

Where

Lecture Theatre 250,
Wilkinson Building

Reminders

- A2b. Progress Presentations: During Week 8 tutorials
- Meeting weekly with your project group members to generate, revise and iterate ideas, to prototype and prepare A2b.
- I will be outside after the lecture for 1:1 consultations

That's the end of today's lecture.

We'll see you in tutorials.

Find your class location here:



THE UNIVERSITY OF
SYDNEY

